

NATURE FOUNDATION MODELS

Building Foundation Models for Natural Systems

Phase 1: Decision Biology
Information-theoretic AI for cellular decision-making under uncertainty

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The Vision

Science Is Still Largely Observational

Despite advances in AI and computation:

- Natural systems remain difficult to predict across biology, chemistry, physics, and environmental sciences
- Scientific knowledge is fragmented across scales, disciplines, and data modalities
- Most scientific discovery workflows remain slow, empirical, and trial-and-error
- We still lack foundational AI systems that can learn, simulate, and reason about the governing dynamics of nature

The Opportunity

Build the foundational AI infrastructure layer for natural systems.

A new class of models that can:

- learn physical and biological dynamics
- model complex adaptive systems
- simulate natural processes under uncertainty
- unify scientific reasoning across domains and scales

What Is a Nature Foundation Model?

A New Category of AI

Nature Foundation Models are:

- foundation models trained on natural system dynamics
- multimodal scientific intelligence systems
- laws-aware models constrained by physics, chemistry, biology, and uncertainty
- world models for living and non-living systems

Goal

Learn the governing information dynamics of nature across scales:

- molecules
- cells
- organisms
- ecosystems
- materials
- environments

The Core Thesis

Nature Is an Information Processing System

Natural systems:

- receive signals
- process noisy information
- make constrained decisions
- adapt under uncertainty

Cells, organisms, and ecosystems can be modeled as probabilistic decision systems governed by:

- information flow
- energy constraints
- physical laws
- stochastic dynamics

The Platform Architecture

The Nature Foundation Models Stack

Natural Data Layer



Scientific Representation Models



Temporal + Dynamics Models



Simulation Infrastructure



Scientific Agents



Design & Discovery Systems

Long-Term Goal

A programmable intelligence layer for science.

Phase 1 — Decision Biology

Decision Biology is the first operational layer of Nature Foundation Models.

Focus

Model how cells:

- process signals
- make decisions
- adapt under uncertainty
- respond to perturbations

Why Cells?

Cells are nature's fundamental decision-making units.

The Problem in Biology

Biology Cannot Reliably Predict Cellular Decisions

Today:

- signaling pathways are treated descriptively
- biological systems are noisy and stochastic
- perturbation outcomes are difficult to predict
- drug discovery remains inefficient
- cellular responses remain poorly quantified

Missing Capability

Reliable modeling of information fidelity inside cells.

Decision Biology

An Information-Theoretic AI Stack for Biology

Decision Biology combines:

- information theory
- foundation models
- probabilistic modeling
- systems biology
- simulation infrastructure
- scientific AI agents

Core Metrics

- Mutual Information
- Channel Capacity
- Signaling Fidelity
- Information Loss
- Perturbation Response